# EduFlow

*Lifeline Learning for Africa's Displaced*

**Product Requirements & Technical Documentation**

Version 1.0 | 2025

# Table of Contents

Table of Contents ... 2
1. Executive Summary ... 4
1.1 Document Purpose ... 4
1.2 Key Metrics at a Glance ... 4
2. Problem Statement ... 5
2.1 The Displacement-Education Crisis in Africa ... 5
2.2 Why Existing EdTech Fails Displaced Learners ... 5
2.3 The Learning Loss Consequences ... 5
3. Solution Overview ... 7
3.1 Core Concept ... 7
3.2 Product Pillars ... 7
3.2.1 Offline Lesson Packs ... 7
3.2.2 SMS/USSD Fallback ... 7
3.2.3 Adaptive Quizzing ... 7
3.2.4 Community Sync ... 7
3.2.5 NGO/Mentor Dashboard ... 7
4. User Research & Target Personas ... 8
4.1 Research Methodology ... 8
4.2 Primary Personas ... 8
Persona 1: Amina — Climate-Displaced Secondary Learner ... 8
Persona 2: Jonas — Conflict-Displaced Urban Learner ... 8
Persona 3: Sister Beatrice — NGO Field Educator ... 8
5. Feature Specifications ... 10
5.1 Feature Priority Matrix (MoSCoW) ... 10
5.2 Detailed Feature Descriptions ... 10
5.2.1 Offline-First Lesson Packs ... 10
5.2.2 Smart Progress Sync ... 11
5.2.3 SMS/USSD Bridge ... 11
5.2.4 Human-Centered UX Design ... 11
5.2.5 Community Sync & Peer Matching ... 11
6. Technical Architecture ... 13
6.1 Architecture Overview ... 13
6.2 Technology Stack ... 13
6.3 Mobile App Architecture ... 13
6.3.1 Flutter Application Structure ... 13

6.3.2 Offline Sync Engine ..... 14
6.4 Backend API Design ..... 14
6.4.1 API Endpoints (REST) ..... 14
6.5 Data Model ..... 14
6.5.1 Core Entities ..... 14
7. Development Roadmap ..... 16
7.1 Hackathon MVP (Week 1) ..... 16
7.2 Phase 2: Post-Hackathon (Months 1–3) ..... 16
7.3 Phase 3: Scale (Months 4–12) ..... 16
8. Impact Framework & Success Metrics ..... 17
8.1 Theory of Change ..... 17
8.2 Key Performance Indicators ..... 17
8.3 Equity Considerations ..... 17
9. Security, Privacy & Ethics ..... 18
9.1 Data Privacy Architecture ..... 18
9.2 Child Safeguarding ..... 18
9.3 Ethical AI Use ..... 18
10. Partnerships & Go-to-Market Strategy ..... 19
10.1 Target NGO & Institutional Partners ..... 19
10.2 Revenue & Sustainability Model ..... 19
11. Risk Assessment & Mitigation ..... 20
12. Hackathon Demo & Submission Plan ..... 21
12.1 Demo Scenario ..... 21
12.2 Submission Checklist ..... 21
13. Appendix ..... 22
A. Glossary ..... 22
B. References ..... 22
C. Team Roles (3-Person Hackathon) ..... 22

# 1. Executive Summary

EduFlow is an offline-first, mobile-first educational platform designed to deliver continuous, high-quality learning to climate- and conflict-displaced youth across sub-Saharan Africa. Built as a progressive Flutter application, EduFlow operates without stable internet connectivity, power infrastructure, or high-end devices — the three most common barriers to education in displacement settings.

The platform combines downloadable lesson packs, SMS/USSD fallback channels, AI-powered adaptive quizzing, and peer community features to create a resilient learning ecosystem that travels with the learner — through floods, forced moves, and fluctuating connectivity.

> **Mission Statement**
> "Education doesn't stop when homes flood — EduFlow puts school in every pocket, ensuring that displacement is never the end of a child's learning journey."

## 1.1 Document Purpose

This document provides comprehensive product, technical, and operational documentation for EduFlow. It is intended for use by:

- Development teams building and maintaining the platform
- NGO and humanitarian partners evaluating or deploying EduFlow
- Investors and grant committees assessing product viability
- Government education ministries reviewing integration pathways
- Hackathon judges and evaluators assessing the Africa Forward submission

## 1.2 Key Metrics at a Glance

| Metric | Target / Value |
|---|---|
| Year-1 Learners (via NGO partners) | 10,000+ |
| Cost per Learner per Year | < $1 USD |
| Minimum Device Requirement | Android 6.0, 1 GB RAM |
| Offline Storage per Lesson Pack | ~50 MB |
| SMS Lesson Delivery Latency | < 30 seconds |
| Cloud Sync Window (on reconnect) | < 2 minutes |
| Target Pass Rate (Adaptive Quizzes) | > 70% after retry |

# 2. Problem Statement

## 2.1 The Displacement-Education Crisis in Africa

Africa hosts the world's fastest-growing displaced populations. According to UNHCR, over 35 million people on the continent are forcibly displaced, with an estimated 40% being children and adolescents of school age. The dual drivers of climate change and armed conflict have compounded this crisis dramatically over the past decade:

- Climate displacement: Floods, droughts, and desertification have displaced millions annually in the Horn of Africa, the Sahel, and Southern Africa.
- Conflict displacement: Ongoing crises in Sudan, the DRC, Ethiopia, Mozambique, and South Sudan continue to uproot families, often repeatedly.
- Compounded vulnerability: Many families experience both climate and conflict displacement, making stable schooling effectively impossible.

## 2.2 Why Existing EdTech Fails Displaced Learners

The majority of existing educational technology platforms are designed for stable, connected, resource-rich environments. They fail displaced learners across several key dimensions:

| Failure Mode | Common EdTech Assumption | Displaced Learner Reality |
|---|---|---|
| Connectivity Dependency | Stable broadband or 4G | Intermittent 2G or none |
| Power Availability | Grid electricity or UPS | Shared solar, no grid |
| Device Access | Personal smartphone/tablet | Shared or borrowed feature phone |
| Account Continuity | Fixed home address & SIM | Frequent SIM/device changes |
| Language & Literacy | Written national language | Low literacy, local dialects |
| Cost | $5–$20/month subscriptions | < $1/year affordable threshold |

## 2.3 The Learning Loss Consequences

The consequences of education interruption for displaced youth are severe and long-term:

- Every year out of school reduces lifetime earnings by an estimated 8–10% (World Bank, 2022).
- Secondary school completion rates in refugee camps remain below 25% across sub-Saharan Africa (UNHCR, 2023).
- Children who miss 2+ consecutive school years are significantly less likely to re-enroll.
- Girls face disproportionate risk — early marriage and domestic labor rapidly fill the vacuum left by school absence.

**The Urgency**

Without intervention, climate change alone could displace an additional 200 million people in Africa by 2050 (World Bank Climate Migration Report). EduFlow addresses the education dimension of this crisis at scale, now.

# 3. Solution Overview

## 3.1 Core Concept

EduFlow is a pocket-sized school — a mobile application that contains everything a displaced secondary learner needs to continue their education, regardless of internet availability, power access, or device quality. It is built on four foundational principles:

1. Offline-First: All critical learning features work without any internet connection.
2. Low-Resource: Designed to run on low-end Android devices with minimal storage and RAM.
3. Human-Centered: Onboarding, content, and UX are tailored to the lived experiences of displaced learners.
4. Resilient Identity: Learner progress is preserved across device changes, SIM swaps, and camp relocations.

## 3.2 Product Pillars

### 3.2.1 Offline Lesson Packs

Downloadable content bundles (approximately 50 MB each) covering core secondary school subjects. Content is pre-cached and playable entirely without network access. Packs are structured around competency levels rather than rigid grade years to accommodate learners who have missed schooling.

### 3.2.2 SMS/USSD Fallback

For learners on basic feature phones with no data access, EduFlow integrates a two-way SMS/USSD channel. Learners can request daily lessons and complete quizzes via text message — ensuring continuity even without a smartphone.

### 3.2.3 Adaptive Quizzing

A lightweight on-device ML model (TensorFlow Lite) tracks quiz performance and dynamically adjusts the difficulty and repetition of topics. Weak areas are automatically reinforced in subsequent sessions without requiring cloud connectivity.

### 3.2.4 Community Sync

When connectivity becomes available, EduFlow matches learners in the same camp, settlement, or region for peer study group formation. This creates social learning bonds that improve retention and reduce isolation.

### 3.2.5 NGO/Mentor Dashboard

A web-based dashboard (Next.js on Vercel) allows NGO staff, camp educators, and remote mentors to monitor learner cohort progress, upload new lesson packs, and identify at-risk learners for targeted outreach.

# 4. User Research & Target Personas

## 4.1 Research Methodology

EduFlow's product design is grounded in desk research, UNHCR field reports, and review of existing humanitarian EdTech deployments including UNICEF's Learning Passport, Kiron Open Higher Education, and Instant Network Schools by Vodafone Foundation. Key qualitative findings informed the feature set and UX decisions.

## 4.2 Primary Personas

### Persona 1: Amina — Climate-Displaced Secondary Learner

| | |
|---|---|
| Age | 15 |
| Location | Dadaab Refugee Complex, Kenya |
| Device | Shared Android (Samsung J2, 2017) |
| Connectivity | Intermittent 2G at camp center, none at shelter |
| Primary Language | Somali; basic English and Swahili |
| Goal | Complete secondary school, become a nurse |
| Pain Points | Missed 18 months of school due to flooding; cannot afford data bundles; shares phone with 4 siblings |

### Persona 2: Jonas — Conflict-Displaced Urban Learner

| | |
|---|---|
| Age | 17 |
| Location | Kampala informal settlement, Uganda |
| Device | Basic Nokia feature phone (SMS only) |
| Connectivity | No smartphone; SMS-capable only |
| Primary Language | Congolese French; learning English |
| Goal | Access vocational training and eventually university |
| Pain Points | No data-capable device; inconsistent access to formal school; works part-time to support family |

### Persona 3: Sister Beatrice — NGO Field Educator

| | |
|---|---|
| Role | Education Officer, IRC Kenya |
| Location | Kakuma Refugee Camp |

| | |
|---|---|
| Goal | Track learner progress; surface at-risk youth |
| Pain Points | Manual paper registers; no visibility on home-based learning; limited staff capacity |

# 5. Feature Specifications

## 5.1 Feature Priority Matrix (MoSCoW)

| Priority | Feature | Phase |
|---|---|---|
| Must Have | Offline lesson packs (Math, Digital Literacy, Languages) | MVP |
| Must Have | Local SQLite progress tracking | MVP |
| Must Have | Basic onboarding & displacement context collection | MVP |
| Must Have | Cloud sync on reconnect | MVP |
| Must Have | SMS/USSD fallback lesson delivery | MVP |
| Should Have | Voice narration for low-literacy users | MVP |
| Should Have | Adaptive quiz engine (TF Lite) | MVP |
| Should Have | NGO web dashboard (basic) | MVP |
| Could Have | Peer community matching | Phase 2 |
| Could Have | Mentor live sessions (low-bandwidth video) | Phase 2 |
| Could Have | Certification and badge system | Phase 2 |
| Won't Have (MVP) | Full LMS, gamification, VR/AR features | Future |

## 5.2 Detailed Feature Descriptions

### 5.2.1 Offline-First Lesson Packs

Lesson packs are self-contained bundles downloadable over any connection type. Each pack includes:

- Structured lesson text with inline illustrations (compressed SVG/WebP)
- Audio narration files (MP3, compressed to < 64kbps for storage efficiency)
- Interactive multiple-choice quizzes with answer feedback
- Glossary and key terms reference
- Competency checkpoints aligned to East African national curricula

Subjects available at MVP launch:

- Mathematics (Secondary Level 1–3): Number systems, algebra, geometry, statistics
- Digital Literacy: Device usage, internet safety, basic productivity tools
- English Language: Reading comprehension, grammar, writing skills
- Swahili Language: Conversational and written Swahili for integration in East Africa

### 5.2.2 Smart Progress Sync

EduFlow uses a dual-layer persistence model to ensure no learning data is lost:

- Local Layer: SQLite database stores all quiz scores, lesson completion timestamps, streak data, and adaptive quiz weights — fully offline.
- Cloud Layer: On any network connection (WiFi or mobile data), a background sync process uploads all pending local events to the Supabase backend.
- Conflict Resolution: Last-write-wins with timestamp-based merging. Device clock skew is handled via server-side timestamp normalization.
- Learner Identity Recovery: If a learner changes devices, their progress is recoverable via SMS OTP linked to their phone number.

### 5.2.3 SMS/USSD Bridge

The SMS/USSD bridge enables learning on the most basic feature phones:

- Inbound: Learner texts a keyword (e.g., "MATH1") to a shortcode number
- Outbound: System responds with a formatted lesson summary and 3-question quiz within 30 seconds
- Quiz Flow: Learner replies with answer code (A, B, C, or D); system responds with result and next question
- Progress Logging: All SMS interactions are logged against the learner's profile for dashboard visibility
- Integration: Twilio or Africa's Talking API for SMS gateway (country-specific shortcodes)

### 5.2.4 Human-Centered UX Design

The onboarding and UX design reflects deep respect for the learner's context:

- Onboarding Question: "Where are you displaced from?" personalizes content tone and peer matching
- Voice Narration Toggle: Activated by default for learners who indicate low reading confidence
- Low-Literacy Mode: Larger fonts, icon-based navigation, simplified language (Flesch-Kincaid Grade 5 max)
- Dark Mode: Reduces battery consumption on OLED screens
- Minimal UI: No ads, no social feeds, no distracting content — learning only

### 5.2.5 Community Sync & Peer Matching

When online, EduFlow's community layer activates:

- Camp/Region Matching: Learners in the same UNHCR camp or GPS-approximate region are matched for study groups
- Study Groups: Text-based async discussion threads organized by subject and lesson

- Mentor Connection: NGO-approved mentors can be linked to learner cohorts for question-answering
- Privacy: No full name or photo required; learners use pseudonyms and camp IDs

# 6. Technical Architecture

## 6.1 Architecture Overview

EduFlow follows a distributed, offline-first architecture. The mobile application is the primary interface and source of truth for learner data. All backend services are additive — they enhance the experience when available but are never required for core learning functionality.

## 6.2 Technology Stack

| Layer | Technology | Justification |
|---|---|---|
| Mobile App | Flutter / Dart | Single codebase for Android and iOS; strong offline support; low APK size footprint |
| Local Storage | SQLite + Hive | SQLite for relational progress data; Hive for fast key-value session state |
| Backend API | Node.js on Render | Simple REST API; cost-effective hosting with free tier for hackathon |
| Database (Cloud) | Supabase (PostgreSQL) | Open-source Firebase alternative; real-time subscriptions; row-level security |
| Web Dashboard | Next.js on Vercel | Server-side rendering for fast load; Vercel free tier; Tailwind CSS |
| SMS Gateway | Africa's Talking / Twilio | Africa-focused SMS pricing; USSD support in East Africa |
| ML / Adaptive Quiz | TensorFlow Lite | On-device inference; no network needed; small model footprint (< 2 MB) |
| CI/CD | GitHub Actions | Automated testing and APK builds on every commit |
| Version Control | GitHub | Team collaboration; PR-based review workflow |

## 6.3 Mobile App Architecture

### 6.3.1 Flutter Application Structure

The Flutter app follows a clean architecture pattern with three layers:

- Presentation Layer: Flutter widgets, BLoC/Provider state management, responsive layouts optimized for 4–6 inch screens
- Domain Layer: Use cases, entities, and repository interfaces — business logic independent of framework
- Data Layer: Local repositories (SQLite/Hive), remote repositories (REST API), and sync orchestration

### 6.3.2 Offline Sync Engine

The sync engine operates as a background Flutter isolate with the following behavior:

5. On app launch, check network status via connectivity_plus package.
6. If offline: queue all write operations (quiz completions, lesson progress) to a local event log.
7. When connectivity is detected: open sync session with backend, replay queued events with timestamps.
8. If sync fails mid-session: mark events as pending and retry on next connectivity window.
9. Notify user of sync status via subtle indicator (no disruptive alerts).

## 6.4 Backend API Design

### 6.4.1 API Endpoints (REST)

| Method | Endpoint | Auth | Description |
|---|---|---|---|
| POST | /auth/register | None | Register learner with phone number; returns JWT |
| POST | /auth/otp-verify | None | Verify SMS OTP; issues session token |
| GET | /lessons/packs | Learner JWT | List available lesson packs with metadata |
| GET | /lessons/packs/:id/download | Learner JWT | Download lesson pack bundle (ZIP) |
| POST | /progress/sync | Learner JWT | Bulk upload local event log; returns server state |
| GET | /progress/:learnerId | NGO JWT | Retrieve learner progress for dashboard |
| POST | /sms/inbound | HMAC Sig | Webhook for incoming SMS events from gateway |
| GET | /community/peers | Learner JWT | Match learner to nearby peers in same region |

## 6.5 Data Model

### 6.5.1 Core Entities

| Entity | Key Fields | Notes |
|---|---|---|
| Learner | id, phone_hash, region, displacement_type, created_at | Phone stored as SHA-256 hash for privacy |
| LessonPack | id, subject, level, language, version, size_mb | Versioned for delta updates |
| Lesson | id, pack_id, sequence, title, content_url, audio_url | Content stored in CDN |
| ProgressEvent | id, learner_id, lesson_id, event_type, score, timestamp | Append-only log for sync replay |
| QuizAdaptation | learner_id, topic_id, weight, last_updated | On-device TF Lite model weights |
| SMSSession | phone, session_state, current_lesson, created_at | Stateful SMS quiz tracking |
| Cohort | id, ngo_id, region, camp_id, members[] | NGO-managed learner groups |

# 7. Development Roadmap

## 7.1 Hackathon MVP (Week 1)

The hackathon MVP is scoped for a 3-person team over 5 working days. Each day targets a shippable vertical slice:

| Day | Focus | Deliverable |
|---|---|---|
| Day 1 | Flutter Scaffolding | App skeleton, offline lesson loader, SQLite schema, basic UI with 2 lesson packs |
| Day 2 | Backend & Auth | Render API deployed, Supabase configured, phone-OTP auth working end-to-end |
| Day 3 | Sync & SMS | Progress sync engine working, SMS webhook live with Africa's Talking sandbox |
| Day 4 | Dashboard & ML | Vercel NGO dashboard with cohort progress view; TF Lite adaptive quiz integrated |
| Day 5 | Polish & Demo | Offline survival test, screen recording, demo video, README, pitch deck |

## 7.2 Phase 2: Post-Hackathon (Months 1–3)

- Expand lesson pack library to include Science, History, and Civic Education
- Launch live peer study group feature with async messaging
- Integrate Bluetooth mesh for offline peer-to-peer lesson sharing (no internet, no cell)
- Conduct field pilots with 3 NGO partners in Kenya and Uganda
- Apply for UNHCR Innovation Fund and Google.org Impact Challenge grants

## 7.3 Phase 3: Scale (Months 4–12)

- Expand to Francophone Africa: French lesson packs, Africa's Talking francophone SMS routing
- Government curriculum alignment: Partner with national education ministries for formal recognition
- Certification pathway: Integrate with Cambridge International and African Union qualification frameworks
- Device partnership: Negotiate pre-installation agreements with low-cost Android OEMs (Tecno, Itel)
- Target: 50,000 active learners across 5 countries by end of Year 1

# 8. Impact Framework & Success Metrics

## 8.1 Theory of Change

EduFlow operates on a clear theory of change: when displaced youth have reliable access to structured learning — regardless of physical location or connectivity — they maintain educational continuity, preserve cognitive development, and increase their likelihood of formal re-enrollment when stability returns.

## 8.2 Key Performance Indicators

| KPI | Year 1 Target | Data Source |
|---|---|---|
| Active learners (monthly) | 10,000 | Supabase dashboard |
| Lesson completion rate | > 60% | Progress sync events |
| 7-day learner retention | > 45% | Cohort analysis |
| Cost per active learner/year | < $1 USD | Infrastructure costs / MAU |
| Quiz pass rate (after adaptive retry) | > 70% | TF Lite outcome log |
| SMS lesson sessions (monthly) | > 5,000 | SMS gateway logs |
| NGO cohorts using dashboard | > 20 | Dashboard login events |
| Learner NPS score | > 50 | In-app survey (quarterly) |

## 8.3 Equity Considerations

EduFlow is designed with equity as a non-negotiable principle:

- Gender parity: Content reviewed for gender-inclusive language and imagery; girl-focused mentorship tracks planned for Phase 2.
- Language access: Swahili and French packs alongside English; community translation program for local dialects in Phase 3.
- Disability inclusion: Voice narration and simplified UI support visual impairments and low literacy; audio-first mode under development.
- No data exploitation: Learner data is never sold or used for commercial advertising; privacy policy aligned with GDPR and Kenya Data Protection Act 2019.

# 9. Security, Privacy & Ethics

## 9.1 Data Privacy Architecture

Given that EduFlow serves a vulnerable population — minors in conflict and displacement settings — privacy and data protection are paramount:

- Phone numbers are stored as SHA-256 hashes; no plain-text PII is stored on servers.
- All API communication uses TLS 1.3 with certificate pinning in the Flutter app.
- Supabase Row Level Security (RLS) ensures learners can only access their own data; NGOs can only access their assigned cohorts.
- JWT tokens expire after 7 days and require silent re-authentication via biometric or PIN.
- No tracking pixels, third-party analytics SDKs, or advertising networks are embedded in the app.

## 9.2 Child Safeguarding

EduFlow implements child safeguarding measures aligned with UNHCR and UNICEF guidelines:

- No public profiles: Learners are never identifiable to other learners by name, photo, or location.
- Peer group vetting: Community matching only occurs within NGO-verified cohorts; no open matchmaking.
- Content moderation: All lesson and community content is pre-moderated; no user-generated content in Phase 1.
- Safeguarding reporting: NGO dashboard includes a flag system for educators to report concerns.

## 9.3 Ethical AI Use

The TensorFlow Lite adaptive quiz engine is designed with algorithmic fairness in mind:

- The model adapts difficulty based on performance, never on demographic characteristics (gender, nationality, displacement type).
- Model weights are reset if a learner explicitly requests a fresh start.
- No behavioral profiling or inferred psychographic data is generated or stored.

# 10. Partnerships & Go-to-Market Strategy

## 10.1 Target NGO & Institutional Partners

| Organization | Engagement Type | Geography |
|---|---|---|
| UNHCR | Learner reach via camp programs; data sharing MOU | Kenya, Uganda, Ethiopia |
| IRC (Int'l Rescue Committee) | Field educator training; dashboard users | East Africa |
| UNICEF Education Section | Curriculum alignment; funding introduction | Regional |
| Kenyan Ministry of Education | Curriculum validation; formal credit pathway | Kenya |
| Africa's Talking | SMS/USSD gateway; startup program access | Pan-Africa |
| Google.org / Google for Education | Infrastructure credits; Impact Challenge grant | Global |

## 10.2 Revenue & Sustainability Model

EduFlow's long-term financial model is designed to keep the learner experience free while generating sustainable revenue:

- NGO SaaS Subscriptions: NGOs pay for the dashboard, cohort management, and reporting features (~$50/cohort/month).
- Government Licensing: National education ministries license EduFlow as a supplementary learning platform for displacement zones.
- Grant Funding: Pursue UNHCR Innovation Fund, Google.org Impact Challenge, Mastercard Foundation, and USAID EdTech grants.
- Device Pre-Installation: Revenue-share agreements with Android OEM partners who pre-install EduFlow on low-cost devices sold in target markets.

# 11. Risk Assessment & Mitigation

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Low smartphone penetration in target camps | High | High | SMS/USSD bridge for feature phones; community device-sharing model |
| Backend outage (Render/Supabase) | Medium | Low | All core features work offline; backend only needed for sync |
| SMS gateway country restrictions | Medium | Medium | Multi-gateway architecture (Africa's Talking + Twilio); country-by-country onboarding |
| NGO data sharing reluctance | Medium | Medium | Privacy-first data model; learner data never shared with NGOs without consent |
| Curriculum misalignment with local standards | Low | High | Competency-based (not grade-based) design; ministry partnership in Year 1 |
| Learner safety in conflict zones | Low | Critical | No GPS data stored; no learner location broadcast; full offline mode |
| Team capacity (3-person hackathon) | High | Medium | Strict scope to MVP; pre-built templates and Supabase reduce build time |

# 12. Hackathon Demo & Submission Plan

## 12.1 Demo Scenario

The live demo for Africa Forward judges will follow this scenario:

10. Open EduFlow on a low-end Android device (Samsung J2, 2017 — no SIM card inserted).
11. Complete offline onboarding: select displacement context, language (Swahili), and subject (Math).
12. Launch an offline Math lesson — demonstrate full interactivity with no network connection.
13. Complete an adaptive quiz; show how the engine flags weak topics for repetition.
14. Enable airplane mode → demonstrate that all progress is saved locally.
15. Disable airplane mode → show background sync completing within 60 seconds.
16. Switch to NGO dashboard on laptop → show learner progress appearing in real time.
17. Send SMS "MATH1" to live shortcode → show quiz delivered via text within 30 seconds.

## 12.2 Submission Checklist

- GitHub repository (public) with README, setup instructions, and demo video link
- APK download link for direct Android installation
- Live Vercel dashboard URL for NGO demo view
- 2-minute pitch video with live demo footage
- This documentation PDF included in repository
- Slide deck (10 slides): Problem, Solution, Demo, Impact, Team, Ask

**Pitch Closing Statement**

"40 million displaced children in Africa deserve more than a promise of education — they deserve a tool that works, wherever they are, with whatever they have. EduFlow is that tool. Built in a week, built to last."

# 13. Appendix

## A. Glossary

| Term | Definition |
|---|---|
| BLoC | Business Logic Component — Flutter state management pattern |
| Hive | Lightweight key-value store for Flutter, used for session caching |
| JWT | JSON Web Token — stateless authentication credential |
| OTP | One-Time Password — used for phone-based authentication |
| RLS | Row Level Security — Supabase database access control |
| SQLite | Embedded relational database used for local progress storage |
| TF Lite | TensorFlow Lite — on-device ML inference framework |
| UNHCR | United Nations High Commissioner for Refugees |
| USSD | Unstructured Supplementary Service Data — menu-based mobile interaction protocol |
| WebP | Modern image format with superior compression vs JPEG/PNG |

## B. References

1. UNHCR Global Trends Report 2023: Forced Displacement Worldwide
2. World Bank (2022): The Human Capital Effects of Education Disruption in Fragile States
3. UNICEF (2023): Education in Emergencies — Sub-Saharan Africa Situation Report
4. Africa's Talking API Documentation — https://developers.africastalking.com
5. Supabase Documentation — https://supabase.com/docs
6. Flutter Offline-First Architecture Guide — https://docs.flutter.dev
7. TensorFlow Lite for Flutter — https://www.tensorflow.org/lite/guide/flutter

## C. Team Roles (3-Person Hackathon)

| Role | Primary Responsibility | Tools |
|---|---|---|
| Lead Dev | Flutter app, SQLite, sync engine | Flutter, Dart, SQLite, TF Lite |
| Backend Dev | Node.js API, Supabase, SMS gateway | Node.js, Supabase, Africa's Talking |

| Role | Primary Responsibility | Tools |
|---|---|---|
| Full-Stack / PM | Next.js dashboard, design, documentation | Next.js, Vercel, Figma, GitHub |